Tilt extraction is a change-from-install calculation: first learn the mounted gravity direction, then report later angle changes relative to that starting position.

1. **Capture the initial calibration window.** After the node is mounted and still, collect

$$C = \{\mathbf{a}_i = [x_i, y_i, z_i]^T\}_{i=1}^{n}.$$

   This quiet window is the zero position.

2. **Estimate gravity and scale.**

$$\mathbf{g}_0 = \frac{1}{n}\sum_{i=1}^{n}\mathbf{a}_i, \qquad s = \|\mathbf{g}_0\|_2, \qquad \hat{\mathbf{d}} = \frac{\mathbf{g}_0}{s}.$$

   Here s is counts per g, and $\hat{\mathbf{d}}$ is the mounted down direction.

3. **Build the local horizontal axes.**

$$\mathbf{p}_x = \mathbf{e}_x - (\mathbf{e}_x^T\hat{\mathbf{d}})\hat{\mathbf{d}}, \qquad \mathbf{u}_x = \frac{\mathbf{p}_x}{\|\mathbf{p}_x\|_2}, \qquad \mathbf{u}_y = \frac{\hat{\mathbf{d}} \times \mathbf{u}_x}{\|\hat{\mathbf{d}} \times \mathbf{u}_x\|_2}.$$

   If sensor x is nearly vertical, seed the plane with sensor y instead.

4. **Project each new x, y, z reading.** For each $\mathbf{a}(t) = [x(t), y(t), z(t)]^T$,

$$v(t) = \frac{\mathbf{a}(t)^T\hat{\mathbf{d}}}{s}, \qquad h_x(t) = \frac{\mathbf{a}(t)^T\mathbf{u}_x}{s}, \qquad h_y(t) = \frac{\mathbf{a}(t)^T\mathbf{u}_y}{s}.$$

   This converts raw counts into vertical and horizontal components in g.

5. **Convert projected acceleration to raw tilt.**

$$\theta_x(t) = \operatorname{atan2}(h_y(t), v(t))\frac{180}{\pi}, \qquad \theta_y(t) = \operatorname{atan2}(h_x(t), v(t))\frac{180}{\pi}.$$

6. **Subtract the initial position.**

$$\Delta\theta_x(t) = \theta_x(t) - \bar{\theta}_x, \qquad \Delta\theta_y(t) = \theta_y(t) - \bar{\theta}_y.$$

   $\bar{\theta}_x$ and $\bar{\theta}_y$ are the calibration-window means, so the final output is tilt change from the mounted starting position.